Definicije

Dobro postavljen problem (PDE + pogoji)

1. Rešitev obstaja
2. Rešitev je ena sama
3. Rešitev je zvezno odvisna od začetnih pogojev (stabilnost)

Karakteristično polje (a, b, c)

Pogoj transverzalnosti $\begin{vmatrix} f'(s) & a(p) \\ g'(s) & b(p) \end{vmatrix} \neq 0$

Začetne trditve

Str:

8 Trditev:

- $U \subseteq \mathbb{R}^n$ odprta okolica ničle
- $f \in \mathcal{C}^2(U)$

Potem je $\Delta f(0) = \lim_{r \searrow 0} \frac{2n}{r^2} \frac{1}{\omega(S_r)} \int_{S_r} (f(x) - f(0))\, d\omega(x)$

$$= \langle f \rangle_{S_r} - f(0)$$

Dokaz:

1. Taylorjeva vrsta za $f = \_\_ + \frac{1}{2} \_\_\_ + o(|x|^2)$
2. integriramo
   - $\int_{S_r} x_j \, d\omega = 0$ (lihost)
   - $\int_{S_r} x_j x_k \, d\omega = 0$ (lihost za $j \neq k$)
   - $\int x_k^2 \, d\omega = \frac{1}{n} \int_{S_r} x_1^2 + \ldots + x_n^2 \, d\omega$ (simetričnost)
3. $\Delta f(0) = \sum \frac{\partial^2 f(0)}{\partial x_i^2}$

$\square$

9 Trditev: Laplaceov operator je invarianten za toge premike in rotacije.

10 KVAZILINEARNE ENAČBE 1. reda

$$a(x,y,u)u_x + b(x,y,u)u_y = c(x,y,u)$$

Ploskev: $(a,b,c)\cdot(u_x, u_y, -1) = 0$

$\vec{r}(x,y) = (x, y, u(x,y))$

normala $= \vec{r}_x \times \vec{r}_y = -(u_x, u_y, -1)$

$(a,b,c)(p)$ podaja tangentno ravnino

$\gamma(t) = (x,y,z)$ $\quad (\gamma(t_0) \subseteq (a,b,c)(p_0),\ \gamma \subseteq \{z = u(x,y)\})$

$\Rightarrow \dot{x} = a \quad \dot{y} = b \quad \dot{z} = c$ $\quad$ - smer odvoda krivulje

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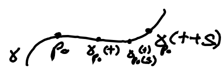
Eksistenčni izrek

- V enakomerno Lipschitzova, zvezna
- $\dot{\xi} = V(\xi) \qquad \xi(0) = \xi_0$

Potem obstaja enolična rešitev v neki okolici začetnega pogoja.

$\dot{\gamma} = V(\gamma) \quad (V = (a,b,c)) \quad \gamma(0) = p_0 = \gamma_{p_0}$

Izkaže se: $\gamma_{p_0}(t+s) = \gamma_{\gamma_{p_0}(s)}(t)$



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Trditev

- $a, b, c \in \mathcal{C}^1(\mathbb{R}^3)$
- $S = \{z = u(x,y)\}$
- $p_0 = (x_0, y_0, z_0) \in S \quad (z_0 = u(x_0, y_0))$

Potem vsaka karakteristiana krivulja za KLE cela leži v S.

Dokaz:

Dokazujemo $\forall t.\ z(t) = u(x(t), y(t))$

oziroma $w(t) = z(t) - u(x(t), y(t)) \equiv 0$

$$\dot{w}(t) = \dot{z} - \dot{x} u_x - \dot{y} u_y =$$
$$= -a(x,y,z) u_x - b(x,y,z) u_y + c(x,y,z) =$$
$$= -a(x, y, w + u(x,y)) u_x - b(x, y, w+u) u_y + \ldots$$
$$= f(t, w)$$

$w(0) = 0$

Ali lahko uporabimo eksistenčni izrek?

Lipschitzovost: zvezno odvedljiva, omejena

$\Rightarrow$ je lipschitzova

Uporabimo eksistenčni izrek:

$w \equiv 0$ reši enačbo. Zaradi enoličnosti

je potem $z(t) = u(x(t), y(t))$

$\square$

Posledica trditve: integralska ploskev je unija karakterističnih funkcij (skozi vsako točko na ploskvi lahko potegnemo karakteristiko)

Posledica:

- $a, b, c \in \mathcal{C}^1(\mathbb{R}^3)$
- $S = \{z = u(x,y)\}$
- $p_0 = (x_0, y_0, z_0) \in S$ $\quad (z_0 = u(x_0, y_0))$

Potem:

1. $p \in S_1 \cap S_2 \Rightarrow$ cela tokovnica skozi p leži v $S_1 \cap S_2$
2. $S_1 \cap S_2$ je krivulja $\Rightarrow$ S_1 in S_2 se NE dotikata in $S_1 \cap S_2$ je integralska krivulja

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Cauchyjev problem za KLE

Iščemo integralsko ploskev, ki vsebuje krivuljo, ki je začetni pogoj.

$R(s,t) = (x(s,t), y(s,t), z(s,t)) = ?$

$$R(s,0) = (f(s), g(s), h(s))$$

Fiksiramo s in rešimo problem

$$au_x + bu_y = c \rightsquigarrow \dot{R}_s = V(R_s) \quad \text{in} \quad R_s(0) = R(s,0)$$

Eksistenčni izrek $\rightsquigarrow$ Imamo $R_s(t)$

Kdaj lahko to sestavimo v ploskev $R(s,t)$?

Poglejmo kdaj $\{R(s,t)\}$ določa graf $\{z = u(x,y)\}$

$x(s,t), y(s,t) \rightsquigarrow$ izrek o inverzni funkciji.

Rabimo pogoj $\frac{\partial(x,y)}{\partial(s,t)}(s_0,0) \neq 0$

$$\begin{vmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{vmatrix}(s_0,0) = \begin{vmatrix} f'(s_0) & a(p_0) \\ g'(s_0) & b(p_0) \end{vmatrix} \neq 0$$

(pogoj transvelzalnosti)

Torej v okolici x_0, y_0 obstaja rešitev.

Pokažimo še enoličnost:

Recimo da u in w rešita problem S_1 za u in S_2 za w se sekata v začetnih pogojih. Torej so vse tokovnice vsebovane v S_1 in v S_2 $\Rightarrow S_1 = S_2$

NELINEARNE ENAČBE 1. REDA

$$F(x,y,u,u_x,u_y)=0 \rightsquigarrow F(x,y,u,p,q)=0$$

Recimo da u, ki reši problem že obstaja.

$S=\{z=u(x,y)\}$ je ploskev z normalo

$$\vec{n}=(u_x,u_y,-1)=(p,q,-1)$$

$\vec{n}(\lambda)=(p(\lambda),q(\lambda),-1)$ družina norm, da

za p in q velja $F(x,y,u,p(\lambda),q(\lambda))=0$

Za Mongojev stožec ustreza enačbama

$$\langle \vec{r}-\vec{r}_0,\vec{n}(\lambda)\rangle=0 \qquad \langle \vec{r}-\vec{r}_0,\vec{n}'(\lambda)\rangle=0$$

$$\vec{n}(\lambda)\times\vec{n}'(\lambda)=(q'(\lambda),-p'(\lambda),pq'(\lambda)-p'q(\lambda))$$

Odvajamo F po λ

$$\frac{\partial}{\partial\lambda}F=p'F_p(x,y,z,p,q)+q'F(x,y,z,p,q)=0$$

$$(p',q')\perp(F_p,F_q)\Rightarrow(F_p,F_q)\parallel(q',-p')$$

$$\Rightarrow(F_p,F_q,\underbrace{pF_p+qF_q}_{})\parallel(q',-p',\underbrace{pq'-p'q}_{})=\vec{r}-\vec{r}_0$$

ohranjanje razmerij $(a,b,pa+qb)$

Parametrizacija stožca

$$r(\lambda)=\vec{r}_0+\mu\cdot(F_p(\lambda),F_q(\lambda),pF_p+qF_q$$

$\gamma(t)=(x,y,z)\in S \Rightarrow \dot{\gamma}(t)\in$ Mongojev stožec

$\Rightarrow$ zahtevamo $\dot{x}=F_p,\ \dot{y}=F_q\ \dot{z}=pF_p+qF_q$

Trditev:

- $\gamma(t) = (x, y, z)$ • $\dot{z} = p\dot{x} + q\dot{y}$

Potem je tangenta na družino krivulj in družino ravnin skozi njene točke določena s p in q

Dokaz: $\langle \dot{\gamma}, (p, q, -1) \rangle = \langle (\dot{x}, \dot{y}, p\dot{x} + q\dot{y}), (p, q, -1) \rangle = 0$

$\square$

$F = 0$ odvajamo po x in y in p in q po t

$$F_x + u_x F_u + p_x F_p + q_x F_q = 0$$
$$p_x F_p = -(F_x + u_x F_u + q_x F_q)$$
$$F_y + u_y F_u + p_y F_p + q_y F_q = 0$$
$$q_y F_q = -(F_y + u_y F_u + p_y F_p)$$
$$\dot{p} = \dot{x} p_x + \dot{y} p_y = p_x F_p + p_y F_q =$$
$$= F_q (p_y - q_x) - (F_x + u_x F_u)$$

Želimo $(u_x)_y = (u_y)_x \rightsquigarrow p_y = q_x$

Torej $\dot{p} = -(F_x + p F_u)$
$$\dot{q} = -(F_y + q F_u)$$

$\boxed{\begin{aligned} &\text{Torej vse skupaj imamo} \\ &\dot{x} = F_p \\ &\dot{y} = F_q \\ &\dot{z} = p F_p + q F_q \\ &\dot{p} = -(F_x + p F_z) \\ &\dot{q} = -(F_y + q F_z) \end{aligned}}$

Trditev:

F je konstantna vzdolž vsake rešitve sistema

$$\dot{x} = F_p \qquad \dot{p} = -(F_x + pF_z)$$
$$\dot{y} = F_q \qquad \dot{q} = -(F_y + qF_z)$$
$$\dot{z} = pF_p + qF_q$$

Dokaz: $\frac{\partial}{\partial t} F(x,y,z,p,q) = \ldots = 0$ $\square$

33 # Cauchyjeva naloga za PDE

Integralska ploskev za $F(x,y,z,p,q)=0$ vsebuje krivuljo $\gamma(s) = (f(s), g(s), h(s))$

5 enačb, 3 pogoji $\rightsquigarrow$ iščemo še ϕ in ψ pogoja

$R(s,t) = (x,y,z,p,q)$

$R(s,0) = (f,g,h,\phi,\psi)$

želimo $z = u(x,y)$, $u_x = p$, $u_y = q \rightsquigarrow \dot{z} = \dot{x}p + \dot{y}q$

$z_s = x_s p + y_s q$

vstavimo $t=0$: $h' = f' \cdot \phi + g' \cdot \psi$

$$\rightsquigarrow \begin{bmatrix} z_s \\ z_t \end{bmatrix} = \underbrace{\begin{bmatrix} x_s & y_s \\ x_t & y_t \end{bmatrix}}_{A} \begin{bmatrix} p \\ q \end{bmatrix}$$

$$\det A \neq 0 \Rightarrow \begin{bmatrix} p \\ q \end{bmatrix} = A^{-1} \begin{bmatrix} z_s \\ z_t \end{bmatrix} = \begin{bmatrix} s_x & t_x \\ s_y & t_y \end{bmatrix} \begin{bmatrix} z_s \\ z_t \end{bmatrix} = \begin{bmatrix} z_x \\ z_y \end{bmatrix}$$

$\gg$

Rešujemo torej sistem:

$$\dot{x} = F_p \qquad \dot{p} = -(F_x + pF_z)$$
$$\dot{y} = F_q \qquad \dot{q} = -(F_y + qF_z)$$
$$\dot{z} = pF_p + qF_q$$
$$h' = \phi p' \cdot \psi q' \qquad \begin{vmatrix} p' & q' \\ F_p & F_q \end{vmatrix} \neq 0$$

Eksistenčni izrek $\rightsquigarrow \exists R(s,t)$. $R(s,0) = (p, q, h, \phi, \psi)$

in (x, y, z) so lokalno graf $u(x,y)$

Ali: x, y, z, p, q rešijo $F(x, y, u, u_x, u_y) = 0$

Preverimo $p = u_x$ in $q = y$

$$\begin{bmatrix} z_s \\ z_t \end{bmatrix} = \begin{bmatrix} x_s & y_s \\ x_t & y_t \end{bmatrix} \begin{bmatrix} z_x \\ z_y \end{bmatrix} = A \begin{bmatrix} u_x \\ u_y \end{bmatrix}$$

če velja $\begin{bmatrix} z_s \\ z_t \end{bmatrix} = A \begin{bmatrix} p \\ q \end{bmatrix}$ in A obrnljiva

$\Rightarrow p = u_x, \ q = u_y$

Torej dokazujemo $z_s = p x_s + q y_s$

37 $B(s,t) = z_s - p x_s - q y_s$

$$\frac{\partial B}{\partial t} = \ldots = \frac{\partial}{\partial s}(z_t - p x_t - q y_t) + p_s x_t + q_s y_t - p_t x_s - q_t y_s =$$
$$= F_p p_s + F_q q_s + (F_x + pF_z) x_s + (F_y + qF_z) y_s =$$
$$= \ldots = -F_z \cdot B$$
$$\rightsquigarrow B(s,t) = B(s,0)\, e^{\int_0^t \ldots dt} \equiv 0$$

LINEARNI DIFERENCIALNI OPERATORJI

$$au_{xx} + 2bu_{xy} + cu_{yy} + du_x + eu_y + fu = g$$

$$M = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \qquad \delta = -\det$$

eliptična $\delta < 0$

hiperbolična $\delta > 0$

parabolična $\delta = 0$

$$\begin{bmatrix} A & B \\ B & C \end{bmatrix} = \begin{bmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \xi_x & \eta_x \\ \xi_y & \eta_y \end{bmatrix}$$

Izrek

- $L(u) = au_{xx} + 2bu_{xy} + cu_{yy} + \ldots + = g$ hiperbolična

Potem $\exists\, \xi, \eta$ koordinate. $w_{\xi\eta} + l_1(w) = \tilde{g}$

44 Dokaz: (ideja)

- želimo $A = C = 0$
- zapišemo v kvadratni enačbi za $\frac{\xi_x}{\xi_y}$
- $a\xi_x + (b + \sqrt{b^2 - ac})\,\xi_y = 0 \rightsquigarrow$ linearna prvega red za $\xi(x,y)$

$\dot{x} = a \qquad \dot{y} = b + \sqrt{b^2 - ac} \qquad \dot{z} = 0$

$\rightsquigarrow \dfrac{\partial y}{\partial x} = \dfrac{\dot{y}}{\dot{x}} \rightsquigarrow y = y(x; C) \rightsquigarrow C = C(x,y) = \xi(x,y)$ $\square$

Izrek (Vmesni)

- $a(x,y), b(x,y)$ zvezno parcialno odvedljivi
- $F = \frac{b}{a}$
- ϕ prvi integral za $y' = F(x,y)$

Potem ϕ reši $au_x + bu_y = 0$

Dokaz:

$$\phi(x, y(x)) \equiv \text{const}$$

obstaja po eksistenčnem izreku za Cauchyjev problem

$$0 = \frac{\partial}{\partial x} \phi(x, y(x)) = \phi_x + y' \cdot \phi_y = \phi_x + F\phi_y$$

$$\Rightarrow a\phi_x + b\phi_y = 0$$

□

Izrek

- $L(u) = au_{xx} + 2bu_{xy} + cu_{yy} + \ldots + = g$ parabolična

Potem $\exists \xi, \eta$ koordinate. $\mathcal{L}w = A\omega_{\xi\xi}$

Dokaz: ($B = C = 0$ ⤳ dovolj $C = 0$)

$\frac{b^2}{a} = c$ ⤳ $C = \frac{1}{a}(a\xi_x + b\xi_y)^2$

Rešujemo $ag_x + bg_y = 0$ linearna 1. reda

$\dot{x} = a \quad \dot{y} = b \quad \dot{z} = 0$ ⤳ $\frac{\partial y}{\partial x} = \frac{\dot{y}}{\dot{x}} = \frac{b}{a}$

Vmesni izrek: Naj bo η prvi integral z $y' = \frac{b}{a}(x,y)$

⤳ $a\eta_x + b\eta_y = 0$ za ξ vzamemo x

$$\begin{vmatrix} 1 & 0 \\ \eta_x & \eta_y \end{vmatrix} = \eta_y$$

če $\eta_y = 0 \Rightarrow \eta_x = 0$

$a\eta_x + b\eta_y = 0$

ker je η netrivialen 1. integral $\Rightarrow \eta_y \neq 0$

□

Izrek

- $L(u) = au_{xx} + 2bu_{xy} + cu_{yy} + \ldots + = g$ eliptična

Potem $\exists\ \xi, \eta$ koordinate. $\mathcal{L}w = A(w_{\xi\xi} + w_{\eta\eta})$

Dokaz:

Želimo $A = C \quad B = 0 \rightsquigarrow A - C + 2iB = 0$

$$A - C + 2iB = \ldots = a\Phi_x^2 + 2b\Phi_x\Phi_y + c\Phi_y^2 \qquad \Phi = \xi + i\eta$$

$$= (a\Phi_x + (b + i\sqrt{-\delta})\Phi_y)(a\Phi_x + (b - i\sqrt{-\delta})\Phi_y)$$

$$\rightsquigarrow \quad a\Phi_x + (b + i\sqrt{-\delta})\Phi_y = 0$$

$$\dot{x} = a \qquad \dot{y} = b + i\sqrt{-\delta} \qquad \frac{\partial x}{\partial y} = \frac{\dot{x}}{\dot{y}} = \frac{b + i\sqrt{-\delta}}{a} =: F$$

$F: \mathbb{R} \to \mathbb{C}$ želimo $F: \mathbb{C} \to \mathbb{C}$ zakaj? ker bemo želeli uporabiti dejstvo, da je Φ holomorfna.

F razširimo na holomorfno z taylorjevo vrsto.

Φ naj bo prvi integral za funkcijo $y' = F(x, y(x))$

in Φ je holomorfna, ker je F holomorfna (se mi zdi).

$\rightsquigarrow \xi_x = \eta_y \quad \xi_y = -\eta_x$

Preverimo, da sta to res koordinate

$$\begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} = \xi_x\eta_y - \xi_y\eta_x = \xi_x^2 + \xi_y^2 = |\nabla\xi|^2 = |\nabla\eta|^2 \neq 0$$

$\square$

56 VALOVNA ENAČBA NA REALNI OSI

$$u_{tt} - c^2 u_{xx} = 0$$

$\delta = c^2 > 0 \rightsquigarrow$ hiperbolične $\rightsquigarrow w_{\xi\eta} = 0 \rightsquigarrow w = F(\xi) + G(\eta)$

$\xi = x - ct \qquad \eta = x + ct$

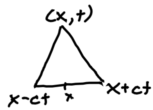
57 Cuchyjev problem $u(x,0) = f(x) \qquad u_t(x,0) = g(x)$

$$F(x) + G(x) = f(x)$$

$$-F(x) + G(x) = \frac{1}{c}\int_0^x g(\xi)\,d\xi + d$$

$$\rightsquigarrow u(t,x) = \frac{1}{2}\left(f(x-ct) + f(x+ct)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} g(\xi)\,d\xi$$

Karakteristični trikotnik



Cauchyjev problem za nehomogeno valovno enačbo

$u_{tt} - c^2 u_{xx} = 0 \quad u(x,0) = f(x) \quad u_t(x,0) = g(x)$

Eksistenčni izrek za nehomogeno valovno enačbo

Rešitev 1. obstaja

2. je enolična

3. je zvezno odvisna od začetnih pogojev

Dokaz:

1. $u_{tt} - c^2 u_{xx} = \varphi \Rightarrow \int\limits_{\Delta} (u_{tt} - c^2 u_{xx}) \, dS = \int\limits_{\Delta} \varphi \, dS$

$$\Rightarrow \int\limits_{\Delta} \varphi \, dS = \int\limits_{\partial\Delta} (u_t \, dt + c^2 u_x \, dx) =$$

$$= \int\limits_{\partial\Delta} (u_t, c^2 u_x) \, d\vec{s} = \ldots =$$

$$= -\int_{x-ct}^{x+ct} g(z) \, dz + 2c\, u(x,t) - c(f(x+ct) + f(x-ct))$$

$$\boxed{\begin{aligned} \Rightarrow u(x,t) &= \frac{f(x+ct) + f(x-ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(z) \, dz + \frac{1}{2} \int\limits_{\Delta} \varphi \, dS \\ &= \langle f \rangle_{\partial I} + t \langle g \rangle_I + \frac{t^2}{2} \langle \varphi \rangle_{\Delta} \\ &\text{Dalembertova formula} \end{aligned}}$$

2. sledi iz 1.

3. Recimo da $\|\varphi_1 - \varphi_2\|_\infty, \|f_1 - f_2\|_\infty, \|g_1 - g_2\|_\infty < \delta$

$$u_1 - u_2 = \langle f_1 - f_2 \rangle_{\partial I} + \langle g_1 - g_2 \rangle_I + \frac{t^2}{2} \langle \varphi_1 - \varphi_2 \rangle \leq$$

$$\leq \delta + t\delta + \frac{t^2}{2}\delta \leq \varepsilon$$

$$\Rightarrow \delta \leq \varepsilon \frac{1}{1 + t + \frac{t^2}{2}}$$

gledamo lokalno za $t \in [0, T]$

$$\Rightarrow \delta = \varepsilon \frac{1}{1 + T + \frac{T^2}{2}} \qquad \square$$

TOPLOTNA ENAČBA (Cauchyjev problem)

$$u_t - \Delta u = 0 \quad \text{na} \quad U \times (0,\infty)$$
$$u = 0 \quad \text{na} \quad \partial U \times [0,\infty)$$
$$u = g \otimes 1 \quad \text{na} \quad U \times \{0\}$$

Separacija spremenljivk $\rightsquigarrow \frac{\Delta X}{X}(x) = \frac{T'}{T} =: -\lambda$

$\rightsquigarrow T(t) = C e^{-\lambda t}$, X je lastni vektor za $-\Delta$

$u(x,0) = X(x) = g(x)$!! g ni nujno lastni vektor za $-\Delta$

Gledamo za $n=1$

$\rightsquigarrow \frac{X''}{X} = -\lambda \rightsquigarrow X = A\cos(\sqrt{\lambda}x) + B\sin(\sqrt{\lambda}x)$

$u=0$ na $\partial U \times [0,\infty) \Rightarrow X \equiv 0$ na $\partial\Omega \qquad U = [a,b]$

$$\begin{bmatrix} \cos(\sqrt{\lambda}a) & \sin(\sqrt{\lambda}a) \\ \cos(\sqrt{\lambda}b) & \sin(\sqrt{\lambda}b) \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = 0$$

predpogoj: $\det(\ldots) = 0 \qquad \det(\ldots) = \sin(\sqrt{\lambda}(b-a))$

$\rightsquigarrow \sqrt{\lambda}(b-a) = k\pi \rightsquigarrow \lambda = \left(\frac{k\pi}{b-a}\right)^2 \quad k \in \mathbb{N}_0$

Poglejmo še $\lambda > 0$

$$-\Delta X = \lambda X \rightsquigarrow \langle -\Delta X, X \rangle = \int_\Omega (-\Delta X) \cdot X \, dV =$$
$$= \int_\Omega \nabla X \cdot \nabla X \, dV + \int_{\partial\Omega} X \cdot \nabla X \, d\vec{S} = \int_\Omega (\nabla X)^2 \geq 0$$
$$= \langle \lambda X, X \rangle = \lambda \|X\|^2 \geq 0 \qquad X \equiv 0 \text{ na } \partial\Omega$$

$\Rightarrow \lambda > 0$

Liho razširimo X čez a $\rightsquigarrow A_k = 0$

$$\sum_{k=1}^{\infty} X_k(x) = \sum B_k \sin\left(\frac{k\pi}{b-a} x\right) = g(x)$$
$$\rightsquigarrow B_k = \frac{2}{b-a} \int_a^b g(y) \sin\left(\frac{k\pi}{b-a} y\right)$$

$$u(t,x) = \sum_{k=1}^{\infty} e^{-\left(\frac{k\pi}{b-a}\right)^2 t} \sin\left(\frac{k\pi}{b-a} y\right) \cdot B_k$$

Ali: $u(x,t)=\sum_{k=1}^{\infty} e^{-\mu_k^2 t} B_k \sin(\mu_k x)$ konvergira

Izrek:

- $I=[a,b] \subseteq \mathbb{R}$
- $v_k \in \mathcal{C}^1(I)$
- $\exists \sigma_0 \in I .\ \sum_{k=0}^{\infty} v_k(\sigma) < \infty$
- $\sum_{k=0}^{\infty} v_k'(\sigma)$ enakomerno konvergira na I

Potem $\frac{\partial}{\partial\sigma} \sum_{k=0}^{\infty} v_k(\sigma) = \sum_{k=0}^{\infty} \frac{\partial}{\partial\sigma} v_k(\sigma)$

Trditev:

- $\sum_{k=1}^{\infty} |c_k| < \infty$

Potem za $u(t,x)=\sum_{k=0}^{\infty} c_k e^{-k^2 t} \sin(kx)$ velja

1) konvergira absolutno na $[0,\pi]\times[0,\infty)=\overline{\Omega}$
2) na $\overline{\Omega}$ določa zvezno funkcijo
3) $Lu=0$ za $L=\partial_t-\partial_{xx}^2$

Dokaz:

1) in 2) $|u(t,x)| \leq \sum_{k=1}^{\infty} |c_k|$. Po Weierstrassovem M-testu velja, da $u(t,x)$ enakomerno konvergira torej je zvezna.

3) $se^{-s} \leq \frac{1}{e}$ za $\forall s \geq 0$ (maksimum funkcije)

$|u_t| = |\sum B_k k^2 e^{-k^2 t} \sin(kx)| \leq |\sum B_k \frac{1}{t} \frac{1}{e} \sin(kx)| \leq \sum B_k$

$|u_{xx}| = |\sum B_k e^{-k^2 t} k^2 \sin(kx)| = |u_t| \leq \sum |B_k|$

$u_t = u_{xx} \Rightarrow Lu = 0$

Torej moramo pokazati, da je $\sum |B_k| < \infty$

$$B_k = \int_0^{\pi} g(y) \sin(ky)\,dy \quad \leftarrow \begin{smallmatrix} a=0 \\ b=\pi \end{smallmatrix} \text{ vpeljava nove spremenljivke}$$

$$B_k = \langle g, \sin(k\cdot)\rangle = \frac{1}{\|\sin(k\cdot)\|} \left\langle g, \frac{\sin(k,\cdot)}{\|\sin(k,\cdot)\|}\right\rangle$$

$$\sum B_k = \frac{2}{\pi}\langle g, \varphi_k\rangle \leq \|g\| \cdot \frac{2}{\pi}$$, ker je φ_k ONS

Katere g lahko razvijemo v Fourierjevo vrsto

$g(a) = g(b) = 0 \rightsquigarrow g$ liho razširimo na $[2a-b, b]$

Če je $g\ \mathcal{C}^1$ potem Fourierjeva vrsta konvergira po točkah.

Nehomogena toplotna enačba

$$u_t - u_{xx} = f \qquad u = 0 \text{ na } \partial U \times [0,\infty) \qquad u = g \otimes 1 \text{ na } U \times \{0\}$$

$$f(x,t) = \sum_{k=1}^{\infty} d_k(t) \sin\left(\frac{k\pi}{b-a}x\right)$$ fourierjeva vrsta po x,

d_k so zato lahko odvisni od t

$\Rightarrow$ Nastavimo $u(t,x) = \sum_{k=0}^{\infty} T(t) \sin(\mu_k x)$

$\rightsquigarrow T' - \mu_k^2 T = d_k \Rightarrow T_k(t) = e^{-\mu_k^2 t} \cdot D_k(t)$, kjer je

$$D_k(t) = \int_0^t d_k(\tau) e^{\mu_k^2 \tau}\,d\tau + D_k(0)$$

$$u(x,0) = \sum T_k(0) \sin(\mu_k x) = g(x) = \sum B_k \sin(\mu_k x)$$

$$T_k(0) = D_k(0) = B_k = \frac{2}{b-a}\int_a^b g(y) \sin(\mu_k y)\,dy$$

$$\Rightarrow u(x,t) = \sum e^{-\mu_k^2 t}\left(\int_0^t d_k(\tau) e^{-\mu_k^2 \tau}\,d\tau + B_k\right) \sin(\mu_k x)$$

Ali je ta rešitev enolična?

$$v = u_1 - u_2 \qquad v_t - v_{xx} = 0 \qquad v = 0 \text{ na } \partial U \times [0,\infty) \qquad v = 0 \text{ na } U \times \{0\}$$

$$E(t) = \int_a^b v^2\,dV \qquad E'(t) = 2\int v_t v\,dV = 2 v_x v \Big|_a^b - 2\int v_x^2\,dV \leq 0$$

$$E(0) = 0 \Rightarrow E \equiv 0 \Rightarrow v \equiv 0$$

HARMONIČNE FUNKCIJE

$\Delta u = 0$

$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right) \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$

$\Delta = 4\partial_z \partial_{\bar{z}}$

Verižno pravilo:

$\frac{\partial}{\partial \bar{z}}(g \circ f) = \left(\frac{\partial}{\partial w} g\right)(f) \cdot \frac{\partial}{\partial \bar{z}} f + \left(\frac{\partial}{\partial \bar{w}} g\right)(f) \cdot \left(\frac{\partial}{\partial \bar{z}} f\right)$

f holomorfna $\Leftrightarrow$ u, v zadoščata Cauchy-Riemanovem sistemu: $u_x = v_y \quad u_y = -v_x$

$\Leftrightarrow \partial_{\bar{z}} f = 0$

f holomorfna $\Rightarrow$ u, v harmonični

Izrek: $\bullet\ u \in \mathcal{C}^2(\Omega)$

Potem u harmonična $\Leftrightarrow$ $u \in LPV$

Dokaz: $u \in LPV \Leftrightarrow \varphi : r \mapsto \langle u \rangle_{K(z_0, r)} = u(z_0)$ konstantna

$\varphi'(s) = \int_0^{2\pi} \frac{d}{ds} u(z_0 + \cos\vartheta, z_0 + \sin\vartheta)\, d\vartheta =$

$= \int_0^{2\pi} (u_x, u_y) \cdot (\sin\vartheta, \cos\vartheta)\, d\vartheta = \int_{\partial K} \nabla u \cdot d\vec{s} =$

$= \int_K \Delta u\, dS$

Torej $u \in LPV \Rightarrow \int_K \Delta u\, dS = 0 \xrightarrow[s \to 0]{} \Delta u(z_0) = 0$

in $\Delta u = 0 \Rightarrow \varphi'(s) = 0 \Rightarrow \lim_{r \to 0} \varphi(r) = \text{konst} = u(z_0)$

$\Rightarrow \varphi(r) = z_0$ $\square$

u subharmonična $(\Delta u \geq 0) \Rightarrow u(a) \leq \langle u \rangle_{\partial K(a, r)}$

Trditev:

- $\Omega^{odpr} \subseteq \mathbb{R}^2$ enostavno povezana
- u realna, harmonična na Ω

Potem $\exists$ hol. funkcija f, da je $\mathrm{Re}(f) = u$

Dokaz:

Iščemo v, da je $\nabla v = \mathrm{div}(v) = (-u_x, u_y)$

Torej želimo, da je $(-u_y, u_x)$ potencialno polje.

Vemo: $\mathrm{Rot}\, F = 0 \Rightarrow F$ potencialno

$\mathrm{Rot}(-u_y, u_x) = u_{xx} + u_{yy} = 0 \Rightarrow \exists v.\ \mathrm{div}(v) = (-u_x, u_y)$

Lema: u harmonična $\wedge$ F holomorfna

$\Rightarrow u \circ F$ harmonična

Dokaz: Vemo: g holomorfna $\Rightarrow \mathrm{Re}(g)$ harmonična

$$u \circ F = (\mathrm{Re}\, \underset{\text{holomorfna}}{f}) \circ F \underset{u \circ F \in \mathbb{R}}{=} \mathrm{Re}(\underset{\text{holomorfna}}{f \circ F})$$

DIRICHLETOV PROBLEM NA $\mathbb{R}^2$

Iščemo $u : \overline{D} \to \mathbb{R}$

$u|_{\partial D} = f$, $\quad u|_{\dot{D}}$ harmonična, u zvezna

Za z^k oziroma $e^{2\pi i k}$ $k \in \mathbb{Z}$ že znamo razširiti

$\rightsquigarrow$ bomo meli za $\sum_{k=-\infty}^{\infty} a_k z^k$ $\qquad r e^{2\pi i k t}$

Izpeljava Poissonovega jedra:

$$f = \sum a_k z^k \qquad a_k = \langle f, e^{i 2\pi k t} \rangle = \int_0^1 f(t) e^{2\pi i k t} dt$$

$$F = \sum a_k z^k \text{ na } \overline{D} : \; F = \sum r^{|k|} e^{2\pi i k t} \int_0^1 f(s) e^{2\pi i k s} ds$$

$$= \sum \int_0^1 r^{|k|} f(s) e^{2\pi i k (t-s)} ds = \int_0^1 f(s) \sum r^{|k|} e^{2\pi i k (t-s)} ds$$

Ali $\sum r^{|k|} e^{2\pi i k s}$ konvergira enakomerno?

$$\sum_{k=0}^{\infty} \left(r e^{2\pi i (t-s)} \right)^k + \sum_{k=1}^{\infty} \left(r e^{-2\pi i s} \right)^k = \ldots = \frac{1-r^2}{|1 - r e^{-2\pi i s}|^2} = \frac{1-|z|^2}{|1-z|^2} =$$

$$= \frac{1-r^2}{1 - 2r\cos(2\pi s) + r^2} = P_r(s) \quad \text{Poissonovo jedro}$$

Približna enota za $r \nearrow 1$

1. $\int_{\partial D} P_r(t)\, dt = 1$ za $\forall r \in [0,1)$
2. $\int |P_r(t)|\, dt < c$ za $\forall r \in [0,1)$
3. $\lim_{r \nearrow 1} \int_{\{\delta < t < \frac{1}{2}\}} |P_r(t)|\, dt = 0$ za $\forall \delta \in (0, \frac{1}{2})$

Izrek:

- $\{P_r(t),\ r\in[0,1)\}$ približna enota za $r\nearrow 1$

Potem za $\forall f: \partial D \to \mathbb{R}$, $f*P_r \xrightarrow{r\nearrow 1} f$ enakomerno na ∂D. $\left(\lim \max_{x\in\partial D} |(f*P_r)(x)-f(x)| = 0\right)$

Dokaz:

$$(f*P_r)(e^{2\pi i t}) - f(e^{2\pi i t}) =$$

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} f(t-s)P_r(s)\,ds - f(t)\cdot \int_{-\frac{1}{2}}^{\frac{1}{2}} P_r(s)\,ds =$$

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} P_r(s)\,(f(t-s)-f(t))\,ds =$$

$$\int_{|s|<\delta} P_r(s)\,(f(t-s)-f(t))\,ds + \int_{|s|\geq\delta} P_r(s)\,f(t-s)-f(t))\,ds$$

$$= \mathrm{I} + \mathrm{II}$$

I: f zvezna na $[\frac{1}{2},\frac{1}{2}] \Rightarrow$ enakomerno zvezna

$\Rightarrow$ δ izberemo tak, da $|f(t-s)-f(t)|<\varepsilon$

$\Rightarrow$ $\mathrm{I} \leq \int_{|s|\delta} P_r(s)\,\varepsilon\,ds \leq \varepsilon \cdot \int_{\partial D} P_r(s)\,ds = \varepsilon$

II: $\mathrm{II} \leq 2\cdot \max_{\partial D} f \cdot \underbrace{\int_{|s|\geq\delta} P_r(s)\,ds}_{<\varepsilon \text{ ker gre proti } 0 \text{ za } r\nearrow 1} \leq 2\max_{\partial D} f \cdot \varepsilon$ za $r\in(r_0,1)$

$\Rightarrow$ Za primeren δ je $\max|(f*P_r)-f| < \varepsilon(1+2\max f)$

$\Rightarrow$ $f*P_r \xrightarrow{r\nearrow 1} f$ enakomerno na ∂D

$\square$

Posledica: $re^{2\pi i t} \mapsto \begin{cases} f(e^{2\pi i t}) ; & r=1 \\ (f*P_r)(e^{2\pi i t}) ; & r<1 \end{cases}$ je zvezna

Izrek:

Poissonovo jedro $\frac{1-r^2}{1-2r\cos(2\pi t)+r^2}$ je približna enota

Dokaz:

Velja: $\int P_r(t)\,dt = \frac{1}{\pi}\arctan\left(\frac{1+r}{1-r}\cdot\tan(\pi t)\right)$

Lastnosti približne enote sledijo iz tega in dejstva $P_r(t)>0$ $\square$

Sklepamo $u=f*P_r$ reši dirichletov problem

$$\Delta u=\Delta\int f(t-s)P_r(s)\,ds \underset{\text{enakomerna konvergenca (zaradi } \partial D \text{ kompakten)}}{=} \int \Delta\Big(f(s)\sum_{k\in\mathbb{Z}} r^{|k|}e^{2\pi i k(t-s)}\Big)\,ds$$

neodvisno od t

$$= \int_0^1 f(s)\,\Delta\sum\ldots\,ds \underset{\text{enakomerna konvergenca } \sum\ldots}{=} \int_0^1 f(s)\sum_{k\geq 0}\Delta(z^k+\bar{z}^k)\,w^{-1}\,dw = 0$$

Želimo še pokazati enoličnost rešitve

Izrek: Princip maksima

- $\Omega\subseteq\mathbb{R}^2$ domočje
- $u\in\mathcal{C}^2$
- $u\in LPV$

Potem če $|u|$ doseže maksimum na $\Omega \Rightarrow u\equiv \text{const.}$

Dokaz:

Recimo, da u doseže maksimum pri z_0

za $\forall r<R$, $u(z_0)=\langle u\rangle_{K(z_0,r)} \xRightarrow{\text{zveznost } u} u\equiv u(z_0)$ na $K(z_0,r)$

ker je to za $\forall r<R$ je $u\equiv u(z_0)$ na $K(z_0,R)$.

Torej je $\{u=u(z_0)\}$ odprta množica in zaprta, ker je praslika točke (ki je zaprta množice). $\square$

Izrek: Dirichletov problem ima največ eno rešitev.

Dokaz:

u_1, u_2 rešitvi. $v = u_1 - u_2$

$\Delta v = 0 \Longrightarrow v \in LPV \Longrightarrow$ velja princip maksima.

$v|_{\partial D} = 0 \Longrightarrow v \equiv 0 \Longrightarrow u_1 = u_2$ $\square$

Izrek:

- $u \in \mathcal{C}_1$

Potem $u \in LPV \Leftrightarrow u$ harmonična

Dokaz:

Na $K = K(z_0, R)$. $u|_{\partial K}$ razširimo na $\tilde{u}$ harm. funkcijo na K.

Potem $u = \tilde{u}$ $\square$

Harmonične konjugiranke al neki

$$\partial D \xrightarrow{f} D \xrightarrow{u} D \xrightarrow{v} \partial D$$

$F(z) = \frac{1+z}{1-z}$ $\quad \mathrm{Re}F = P_r(z) = \frac{1-|z|}{|1-z|^2}$

Definirajmo $Q = \mathrm{Im}F = \frac{2\,\mathrm{Im}z}{|1-z|^2} = \frac{2r\sin(2\pi t)}{1-2r\cos(2\pi t)+r^2}$

Q_r ni približna enota

Želimo: $v = f * Q_r \qquad u = f * P_r$

$$u + iv = \int_0^1 (f(s)P(t-s) + i f(s)\,Q(t-s))\,ds =$$

$$= \int_0^1 f(s)F(t-s)\,ds = f * F$$

$f*F$ je holomorfna, ker $\frac{\partial}{\partial \bar z}\int f(w)F(zw^{-1})\,dw =$

$= \int f(w)\frac{\partial}{\partial \bar z}F(zw^{-1})\,dw = \int f(w)\cdot 0\,dw = 0$

kompozitum holomorfnih

Torej mora $f*F$ enakomerno konvergirati za $\forall r$

To velja za $\forall r < 1$

Želimo najti Q_1, ker bo $g = f * Q_1$

$Q_1(t) := \lim_{r \nearrow 1} P_r(t) = \begin{cases} 0 & ; t=0 \\ \cot(\pi t) & t \neq 0 \end{cases}$ $\quad$ ker $\frac{\sin x}{1-\cos x} = \cot\frac{x}{2}$

Izrek:

- f odvedljiva

Potem $\lim\limits_{\delta \searrow 0} \int\limits_{\{\delta < s < \frac{1}{2}\}} f(t-s)\cot(\pi s)\,ds$ obstaja

Dokaz:

$$\int\limits_{\{\delta < s < \frac{1}{2}\}} f(t-s)\cot(\pi s)\,ds = \int_\delta^{\frac{1}{2}} f(t-s)\cot(\pi s) - \int_\delta^{\frac{1}{2}} f(t+s)\cot(\pi s)\,ds$$

$$= \int_\delta^{\frac{1}{2}} \cot(\pi s)\big(f(t-s) - f(t+s)\big)\,ds =$$

$$= \int_\delta^{\frac{1}{2}} \cot(\pi s)\cdot s\left(\frac{f(t-s)-f(t)}{s} - \frac{f(t+s)-f(t)}{s}\right)ds$$

$$= \int_\delta^{\frac{1}{2}} \underbrace{\cot(\pi s)\cdot s}_{\text{obstaja v } 0}\,\underbrace{\big(-f'(\xi_s) - f'(\eta_s)\big)}_{\text{obstaja v } 0}\,ds$$

Hilbertova transformiranka $H_f(t) = \lim\limits_{\delta \searrow 0} \int\limits_{\{\delta < |s| < \frac{1}{2}\}} f(t-s)\cot(\pi s)\,ds$

Trditev: $\lim_{r \nearrow 1} (f * Q_r)(t) = H_f(t)$

Dokaz: $\delta = 1-r$

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} Q_r(s) f(t-s)\, ds = \int_{\{s<\delta\}} Q_r(s) f(t-s)\, ds + \int_{\{s\geq\delta\}} Q_r(s) f(t-s)\, ds$$
$$= \quad I \quad + II$$

I: $f(t) \int_{\{s<\delta\}} Q_r(s)\, ds = 0$ (lihost)

$$I = \int_{\{s<\delta\}} Q_r(s)(f(t-s) - f(t))\, ds = \int_{\{s<\delta\}} Q_r(s)\, s\, \underbrace{\frac{f(t-s)-f(t)}{s}}_{\leq C_t} ds \leq$$
$$\leq 2\delta \cdot C_t \cdot \sup(s Q_r(s))$$
$$s Q_r(s) = w \frac{2 \operatorname{Im} w}{|1-w|^2} = \frac{2s \sin s}{|1-w|^2} \leq \frac{2 \sin^2 s}{|1-w|^2} =$$
$$= \left(\sqrt{2} \frac{\operatorname{Im} w}{|1-w|}\right)^2 = \left(\sqrt{2} \frac{|w-\bar{w}|}{|1-w|}\right)^2 =$$
$$4 \left| \frac{1-w}{|1-w|} - \frac{1-\bar{w}}{|1-\bar{w}|} \right|^2 \leq 8$$
$$\Rightarrow I \leq 16 \delta C_t \xrightarrow{\delta \searrow 0} 0$$

Želimo torej $\lim_{\delta \searrow 0} II = H_f(t)$

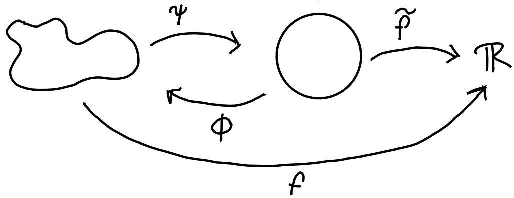
$$II - H_f(s) = \int_{\{\delta \leq \frac{1}{2}\}} (Q_1(s) - Q_r(s)) f(t-s)\, ds$$
$$Q_1(s) - Q_r(s) = \frac{\sin(2\pi s)}{1-\cos(2\pi s)} - \frac{2r \sin(2\pi s)}{1-2r\cos(2\pi s)+r^2} =$$
$$= \sin x \frac{1-2r\cos x + r^2 - 2r + 2r\cos x}{(1-\cos x)(1-2r\cos x + r^2)} = \frac{\sin x \cdot (1-2r+r^2)}{\cdots}$$
$$= (1-r)^2 \frac{\sin x}{1-\cos^2 x}(1+\cos x) \frac{1}{1-2r\cos x + r^2} =$$
$$= (1-r)^2 \frac{1+\cos x}{\sin x} P_r(s) \frac{1}{1-r^2} = \frac{1-r}{1+r} \frac{1+\cos x}{\sin x} P_r(s)$$
$$\Rightarrow \left| \lim_{r \nearrow 1} II - H_f(s) \right| = 0$$

$\square$

Tako smo našli našo konjugiranko za f

$$g = H_f(t) = \lim_{r \nearrow 1} (f * Q_r)(t)$$

DIRICHLETOV PROBLEM NA POLJUBNIH DOMENAH



- imamo f na $\partial\Omega$
- $f(x) = \tilde{f}(\psi(x))$
- $u(x) = \tilde{u}(\psi(x)) = (\tilde{f} * P_r)\,\psi(x)$

$$u(z) = u(\phi(\tilde{z})) = (u\circ\phi)(\tilde{z}) = \frac{1}{2\pi}\int_{\partial D} (f\circ\phi)(\tilde{w})\, P(z\,\overline{\tilde{w}})\, d\tilde{w} =$$

$$= \int_{\partial D} f(\phi(\tilde{w}))\, P(z\,\overline{\tilde{w}})\, d\tilde{w} \underset{w=\phi(\tilde{w})}{=} \frac{1}{2\pi}\int_{\partial\Omega} f(w)\, P(z\,\overline{\psi(w)})\, |\partial\psi(w)|\, dw$$

$$P_\Omega(z,w) = \frac{1}{2\pi} P_r(z\,\overline{\psi(w)})\, |\partial\psi(w)|$$

Torej imamo rešitev $u(z) = \int_{\partial\Omega} f(w)\, P_\Omega(z,w)\, dw$

GREENOVE FUNKCIJE

Želimo rešiti $Lu=f$ $\left(L=\sum_{i=1}^{n}\frac{\partial^{m_i}}{\partial x_i^{m_i}}\right)$

$Lw=\delta \quad / * f$

$L(w*f)=(Lw)*f=\delta*f=f$

Torej če za L znamo rešiti $Lw=\delta$, bomo znali rešiti $Lu=f$

Želimo odgovor za $L=\Delta$

$\Delta u=f$ na Ω

$u=g$ na $\partial\Omega$

u zvezna na $\overline{\Omega}$

Nastavek za w: $w(z)=\Psi(|x-\xi|)=\Psi(R)$

$\Delta\Psi=\Psi''+\frac{n-1}{R}\Psi'$

$\Psi'=\eta \qquad \eta'=-\frac{n-1}{R}\eta+\delta$

$\eta^{-1}d\eta=-(n-1)R^{-1}dr$

$\ln\eta=(1-n)\ln R+c$

$\eta=cR^{1-n} \quad \rightsquigarrow \quad \Psi(r)=\begin{cases}c\ln R+D & ;n=2\\ \frac{c}{2-n}R^{2-n}+D & ;n>2\end{cases}$

Zato za $n=2$ vzmemo $K(z,\xi)=\frac{1}{2\pi}\log|z-\xi|$

Izrek: Greenova reprezentacijska formula

- $u \in \mathcal{C}^2(\bar{M})$ • $z \in M$

Velja:

$$u(z) = \int_{\partial M} (u \cdot \nabla K - \nabla u \cdot K)(z,\cdot)\, d\vec{s} + \iint_M (K \cdot \Delta u)(z,\cdot)\, dS$$

Dokaz:

- Vzemimo $\Omega = M - \overline{K(z,\varepsilon)}$
- Greenova 3. identiteta:

$$\int_{\partial\Omega} u \nabla K - \nabla u \cdot K)\, d\vec{s} = \iint_\Omega (u \cdot \overset{\overset{0}{\parallel}}{\Delta K} - \Delta u \cdot K)\, dS =$$

$$= -\iint_\Omega \Delta u \cdot K\, dS \xrightarrow[\varepsilon \searrow 0]{} \int_M K \cdot \Delta u\, dS$$

$$\Rightarrow \int_M K \cdot \Delta u\, dS = \int_{\partial M} (u \nabla K - \nabla u \cdot K) - \lim_{\varepsilon \searrow 0} \int_{K(z,\varepsilon)} (u \nabla K - \nabla u K)\, d\vec{s}$$

$$\int_{\partial K(z,\varepsilon)} \nabla u \cdot K\, ds = \frac{1}{2\pi} \int_{\partial K} \nabla u \cdot \log \varepsilon\, d\vec{s} \le \frac{1}{2\pi} \log \varepsilon \max \nabla u \cdot 2\pi\varepsilon \xrightarrow[\varepsilon \searrow 0]{} 0$$

$\int_{\partial K} u \cdot \nabla K\, d\vec{s}$:

$$\nabla(\log|z-\zeta|) = \frac{1}{2} \nabla(\log((z_1-\zeta_1)^2 + (z_2-\zeta_2)^2)) =$$

$$= \frac{1}{2} \frac{1}{|z-\zeta|^2} (2(z_1-\zeta_1), 2(z_2,\zeta_2)) = \frac{1}{\varepsilon^2}(z-\zeta)$$

$$\Rightarrow \int_{\partial K} u \nabla K\, d\vec{s} = \frac{1}{\varepsilon^2}(z-\varepsilon) \frac{1}{2\pi} \int_{\partial K} u\, d\vec{s} \longrightarrow \langle u \rangle_{\partial K} \longrightarrow u(z)$$

$$\Rightarrow \int_M K \Delta u\, dS = \int_{\partial M} (u \cdot \nabla K - \nabla u \cdot K)\, d\vec{s} - u(z) \qquad \square$$

$$\Gamma(f)(z) = \iint_M f(\zeta) \cdot K(z,\zeta)\, dS(\zeta)$$

$$\Gamma(\Delta u) = \iint_M \Delta u \cdot K = u - \int_{\partial M} (u \cdot \nabla K - \nabla u \cdot K)\, ds$$

Če je $u \in \mathcal{C}_c^2(\mathbb{R})$ potem je $u = \Delta u = 0$ na ∂M

$\Rightarrow \Gamma(\Delta u) = u$

Torej imamo rešitev na $\mathcal{C}_c^\infty$ za $\Delta u = f$

$$u = \Gamma(\Delta u) = \Gamma(f)$$

Imamo torej formulo

$$u(z) = \int_{\partial M} (u \cdot \nabla K - \nabla u \cdot K)(z,\cdot)\, d\vec{s} + \iint_M (K \cdot \Delta u)(z,\cdot)\, dS,$$

ampak ne vemo koliko je ∇u, torej želimo, da je $K=0$ na ∂M, da se znebimo tega člena

Poskus: $G = K + h$ za nek harmoničen h

Poglejmo za poljuben h:

$$\int_{\partial M} (u \nabla G - \nabla u G)\, d\vec{s} + \iint_M G \cdot \Delta u\, dS =$$

$$= \int_{\partial M} (u \nabla K - \nabla u K)\, d\vec{s} + \iint_M K \Delta u\, dS + \int_{\partial M} u \nabla h - \nabla u h + \iint_M h \Delta u\, dS =$$

$$= u(z) + \iint (u \Delta h - \Delta u h)\, dS + \iint_M h \Delta u\, dS = u(z)$$

$$h(z,\zeta) = \int_{\partial M} K(w,\zeta) P_M(z,w)\, dw = \frac{1}{2\pi} \int_{\partial M} \log|w-\zeta|\, P_M(z,w)\, dw$$

$$u(z) = \int_{\partial M} u \nabla G\, d\vec{s} + \iint_M G \cdot \Delta u\, dS$$

$G = K + h$ je **greenova funkcija** domene M za Laplacov operator

Rešitev dirichletovega problema

- $\Delta u = f$ na M
- $u = g$ na ∂M
- u zvezna na $\overline{M}$

$$\Delta u = f,\ u = g \text{ na } \partial M \Longrightarrow u = \int_{\partial M} g \cdot \nabla G\, d\vec{s} + \iint_M G \cdot f$$

Poissonovo jedro definiramo kot $P_{M,z}(\zeta) = \frac{\partial}{\partial \vec{n}} G(z,\zeta)$

odvod po zunanji normali

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SCHWARTZOVE FUNKCIJE

$$\rho_{\alpha,\beta}(f) = \sup |x^\alpha \cdot \partial^\beta f| = \sup \left|\prod_{i=1}^{n} x^{\alpha_i} \cdot \frac{\partial^{|\beta|} f}{\partial x_1^{\beta_1} \cdots \partial x_n^{\beta_n}}\right|$$

$$\rho_{\alpha,\beta}(f+g) = \sup |x^\alpha \cdot (\partial^\beta f + \partial^\beta g)| \leq \rho_{\alpha,\beta}(f) + \rho_{\alpha,\beta}(g)$$

$$\rho_{\alpha,\beta}(\lambda f) = \lambda \cdot \rho_{\alpha,\beta}(f)$$

Schwartzov razred $\mathcal{S}(\mathbb{R}^n) = \{ f \in \mathcal{C}(\mathbb{R}^n);\ \rho_{\alpha,\beta}(f) < \infty \}$

$\mathcal{C}_c \subseteq \mathcal{S} \subseteq \mathcal{C}^\infty$

Trditev: • $f \in \mathcal{C}^\infty$.

$f \in \mathcal{S} \iff \forall \beta \in \mathbb{N}_0^n.\ \forall n \in \mathbb{N}.\ \sup (1+|x|)^N |\partial^\beta f| < \infty$

Dokaz: (približno)

$$\sup ((1+|x|)^N |\partial^\beta f| = \sup \left(\sum_{k=1}^{N} a_k |x|^k |\partial^\beta f| \right)$$

$(\Rightarrow)$ $\sup |x^\alpha \cdot \partial^\beta f| \leq \sup |x|^{|\alpha|} |\partial^\beta f| \leq \sup (1+|x|)^{|\alpha|} |\partial^\beta f| < \infty$

$(\Leftarrow)$ $\sup (\sum a_k |x|^k |\partial^\beta f|) \leq C \cdot \sup |x|^N |\partial^\beta f| \leq C \cdot \rho_{(N,N,\dots N),\beta}(f) < \infty$

Definiramo: $f_k \longrightarrow f$ v $\mathcal{S} \iff \forall \alpha,\beta.\ \rho_{\alpha,\beta}(f_k - f) \xrightarrow[k\to\infty]{} 0$

FOURIERJEVA TRANSFORMACIJA

$\mathcal{F}: f \mapsto \hat{f} \qquad \hat{f}(\xi) = \int_{\mathbb{R}^n} f(z) e^{-2\pi i \langle z, \xi \rangle} dz$

Izrek:

- $f, g \in \mathcal{S}(\mathbb{R}^n)$ • $y \in \mathbb{R}^n$ • $\alpha \in \mathbb{N}_0^n$

Potem velja

1) $\|\hat{f}\|_\infty \leq \|f\|_1 \quad \left(\sup|\hat{f}| \leq \int |f|\right)$

8) $\widehat{\partial^\alpha f}(\xi) = (2\pi i \xi)^\alpha \hat{f}(\xi)$

9) $\partial^\alpha \hat{f} = \widehat{(-2\pi i x)^\alpha f}(\xi)$

10) $\hat{f} \in \mathcal{S}$

11) $\widehat{f * g} = \hat{f} \cdot \hat{g}$

Dokaz:

1)
$$\|\hat{f}\|_\infty = \sup \left| \int f(\xi) e^{2\pi i \langle z, \xi \rangle} d\xi \right| \leq \sup \int |f(\xi)| d\xi = \sup \|f\|_1 = \|f\|_1$$

8) Indukcija po $|\alpha|$
$$\int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} \frac{\partial}{\partial x_i} \partial^{\alpha'} f(\xi) e^{-2\pi i \langle z, \xi \rangle} = \int_{\mathbb{R}^{n-1}} \partial^\alpha f e^{-2\pi i \langle z, \xi \rangle} \Big|_{\mathbb{R}^n} +$$
$$+ \int_{\mathbb{R}^n} \partial^{\alpha'} f(\xi) \cdot 2\pi i \, \xi_i \, e^{-2\pi i \langle z, \xi \rangle} dz =$$
$$= 2\pi i \xi_i \cdot (2\pi i \xi)^{\alpha'} \hat{f}(\xi) = (2\pi i \xi)^\alpha \hat{f}(\xi)$$

Baza: $\alpha = (0, 0, \ldots, 0)$: $\hat{f}(\xi) = (2\pi i \xi)^0 \hat{f}(\xi)$

9) $\partial^\alpha \hat{f} = \frac{\partial}{\partial x_i} \partial^{\alpha'} \hat{f} = \frac{\partial}{\partial x_i} \widehat{(-2\pi i x)^{\alpha'} f}(\xi) =$
$$= \frac{\partial}{\partial \xi_i} \int (-2\pi i x)^{\alpha'} f(x) e^{-2\pi i \langle x, \xi \rangle} dx =$$
$$= \int (-2\pi i x)^\alpha f(x) e^{-2\pi i \langle x, \xi \rangle} dx = \widehat{(-2\pi i x)^\alpha f}(\xi)$$

Baza: $\hat{f} = \hat{f}$

10) $\rho_{\alpha, \beta}(\hat{f}) = \sup x^\alpha \cdot \partial^\beta \int f(\xi) e^{-2\pi i \langle x, \xi \rangle} d\xi =$
$$= \sup x^\alpha \int (-2\pi i \xi)^\beta f(\xi) e^{-2\pi i \langle x, \xi \rangle} d\xi =$$
$$= \max\{\sup_K \ldots, \sup_{\mathbb{R} \setminus K} \ldots\}$$
$$\leq \max\left\{C, \int (-2\pi i \xi)^{\beta + \alpha} \frac{1}{(-2\pi i)^\alpha} f(\xi) e^{-2\pi i \langle x, \xi \rangle} d\xi\right\} \leq$$
$$= \max\left\{C, \frac{1}{(2\pi i)^\alpha} \partial^{\alpha + \beta} f\right\} < \infty$$

11) $\widehat{f * g} = \iint f(\xi) g(z - \xi) d\xi \cdot e^{-2\pi i \langle x, z \rangle} dz =$
$$= \int d\xi \int f(\xi) g(z - \xi) e^{-2\pi i \langle x, z - \xi \rangle} e^{-2\pi i \langle x, \xi \rangle} dz =$$
$$= \int \hat{g}(x) \cdot f(\xi) e^{-2\pi i \langle x, \xi \rangle} d\xi = \hat{g} \cdot \hat{f}$$

$\check{f}(x) = \hat{f}(-x)$ inverzna fourierova transformacija

Izrek:

- $f, g \in \mathcal{S}(\mathbb{R}^n)$

Velja:

1) $\int f \hat{g} = \int \hat{f} g$
2) $(\hat{f})^\vee = (\check{f})^\wedge = f$
3) $\langle f, g \rangle = \langle \hat{f}, \hat{g} \rangle$ za $\langle \varphi, \psi \rangle = \int \varphi \cdot \overline{\psi}$
4) $\|f\|_2 = \|\hat{f}\|_2 = \|\check{f}\|_2$ Plancherelova identiteta
5) $\int f g = \int \hat{f} \check{g}$

Izrek: $\mathcal{F} : \mathcal{S} \to \mathcal{S}$ je homeomorfizem

Bijektivnost: $\mathcal{F}(f) = \hat{f}$ $\leadsto$ $g = \mathcal{F}(\check{g})$

Zveznost: Ali za poljubno zaporedje $(f)_k \to f$

velja $\mathcal{F}(f_k) \to \mathcal{F}(f)$?

Recimo da $f_k \to f$ (torej $\rho_{\alpha,\beta}(f_k - f) \to 0$)

$$g = f_k - f \quad \hat{g} = \widehat{f_k - f} = \hat{f_k} - \hat{f}$$

$$\rho_{\alpha,\beta}(\hat{g}) = \| x^\alpha \partial^\beta \hat{g}\|_\infty = \| x^\alpha \widehat{(-2\pi i \xi)^\beta g_k}(x)\|_\infty =$$

$$\leq \| x^\alpha (-2\pi)^{|\beta|} \cdot \widehat{\xi^\beta g_k}(x)\|_\infty =$$

$$= \|(-2\pi i x)^\alpha (-2\pi)^{\beta - \alpha} \widehat{\xi^\beta g_k}(x)\| =$$

$$= \|(-2\pi)^{\beta - \alpha} \widehat{\partial^\alpha(\xi^\beta g_k)}(x)\|_\infty$$

$$\leq (-2\pi)^{\beta-\alpha} \|\partial^\alpha(\xi^\beta g_k)\|_1 \leq C_n \sum_{|\gamma| \leq n+1} \rho_{\gamma,\alpha}(x^\beta g_k) =$$

$$= C_n \sum \sum x^\delta \partial^\varepsilon g_k \leq C_n \sum \rho_{a,b}(g_k) \xrightarrow{k \to \infty} 0$$

Zveznost inverza:

$$\mathcal{F}(f) = \hat{f} \quad \mathcal{F}^{-1}(f) = \check{f} = \hat{f}(-x) = \hat{f} \circ (x \mapsto -x)$$

kompozitum dveh zveznih funkcij je zvezna $\square$

Rieman-Lebesguova lema

$\forall f \in L^1(\mathbb{R}).\ \lim_{|\xi|\to\infty} |\hat{f}(\xi)| = 0$

Dokaz:

Vemo: $\mathcal{C}_c^\infty \subseteq \mathcal{S} \subseteq L^1$ in $\mathcal{C}^\infty$ gosta v L^1

Zato je $\mathcal{S}$ gosta v L^1 (tudi v L^2)

$f \in L^1$ poljuben in $g \in \mathcal{S}$ tak, da velja $\|f-g\|_{L^1} < \varepsilon$

$$|\hat{f}(\xi)| \leq |\hat{f}(\xi) - \hat{g}(\xi)| + |\hat{g}(\xi)| \leq |\widehat{f-g}(\xi)| + \varepsilon \leq 2\varepsilon$$

zadosti velik ξ

$$\|f-g\|_{L^1} = \int |f-g|\, d\xi \geq \left| \int (f-g) e^{-2\pi i \langle x, \xi \rangle} d\xi \right| = |\widehat{(f-g)}(\xi)|$$

$\square$

RAZŠIRITEV $\mathcal{F}$ NA $\mathcal{L}^p$ $1<p<2$

Želimo: $f = f_1 + f_2$; $f_1 \in L_1$, $f_2 \in L_2$

$f_1 = f \cdot \chi_{\{|f| \geq 1\}}$, $f_2 = f \cdot \chi_{\{|f| \leq 1\}}$ $\Rightarrow f \leq f^p$

Izrek: Housdorf-Young

- $p \in [1,2]$ • $f \in L^p$ • $\frac{1}{p} + \frac{1}{q} = 1$

Velja $\|\hat{f}\|_q \leq \|f\|_p$

Izrek: Beckner

- $p \in [1,2]$ • $f \in L^p$ • $\frac{1}{p} + \frac{1}{q} = 1$

Velja točna ocena $\|\hat{f}\|_q \leq \left(p^{\frac{1}{p}} \cdot q^{-\frac{1}{q}}\right)^{\frac{n}{2}} \|f\|_p$

Ekstremalne so Gaussove funkcije